
LOOK BEFORE YOU SLEW: INFORMATION-GATED DYNAMIC TASKING FOR AGILE EARTH OBSERVATION

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ABSTRACT

Earth-observing satellites typically execute schedules fixed before overflight, so clouds and transient events can render planned images low-value while better opportunities go unused. A secondary body-fixed sensor can inspect upcoming primary-imager accesses before acquisition, but each lookahead consumes maneuver time and may displace imaging. We therefore observe only when expected schedule improvement exceeds this opportunity cost. After classification, a local scheduler removes cloudy tasks and inserts clear alternatives within the schedule fragment bounded by neighboring committed observations. We derive body-fixed field-of-view and agility constraints, a two-anchor renewal estimator of insertion value, and a monotone break-even approximation. The resulting receding-horizon policy searches feasible pitches, initiates each lookahead at the latest feasible time, updates candidate utilities, and repairs the local schedule. On sampled IID repair windows, the two-anchor estimate closely matches Monte Carlo insertion value, while the cheaper break-even rule is conservative. In paired 24 h scheduling trials, the value-based policies close about 78 %–79 % of the conventional-to-omniscient gap under both IID cloud truth and realistic Binary Cloud Mask fields, while using about 40 % fewer lookahead actions per scheduled access than a greedy lookahead baseline. A case study recovers most achievable single-pass repair benefit with only a few targeted lookaheads.

Keywords: dynamic tasking, Earth observation, spacecraft scheduling, onboard autonomy, agile satellites

🔗 https://github.com/Shreeyam/dt_sim

1 Introduction

Modern Earth-observing (EO) satellites still operate with limited autonomy: ground-optimized schedules are uplinked and executed with little response to uncertainty. This mode of operation works effectively when target utility is known or can be accurately forecast, but can waste opportunities when transient events such as wildfires, floods, or cloud cover change the utility of targets. Cloud cover is the most common example: clouds obscure approximately 66% of Earth’s surface at any one time [1], obscuring a target when alternatives may be clear.

Dynamic tasking addresses this limitation by repairing the onboard schedule as new utility information becomes available. That information may come from a secondary sensor, another spacecraft, or an external meteorological product. We consider a secondary body-fixed, wide-field instrument paired with an onboard classifier that updates access utilities and a scheduler that repairs the affected schedule fragment (Figure 1).

Acquiring new information has its own cost. A lookahead may reveal viable alternative images, but it consumes maneuver and inference time and may displace imaging already in the schedule. Body-fixed lookahead is therefore a value-of-information problem: the spacecraft is not simply choosing its next image, but deciding whether an information-gathering maneuver will improve the downstream schedule enough to justify its cost.

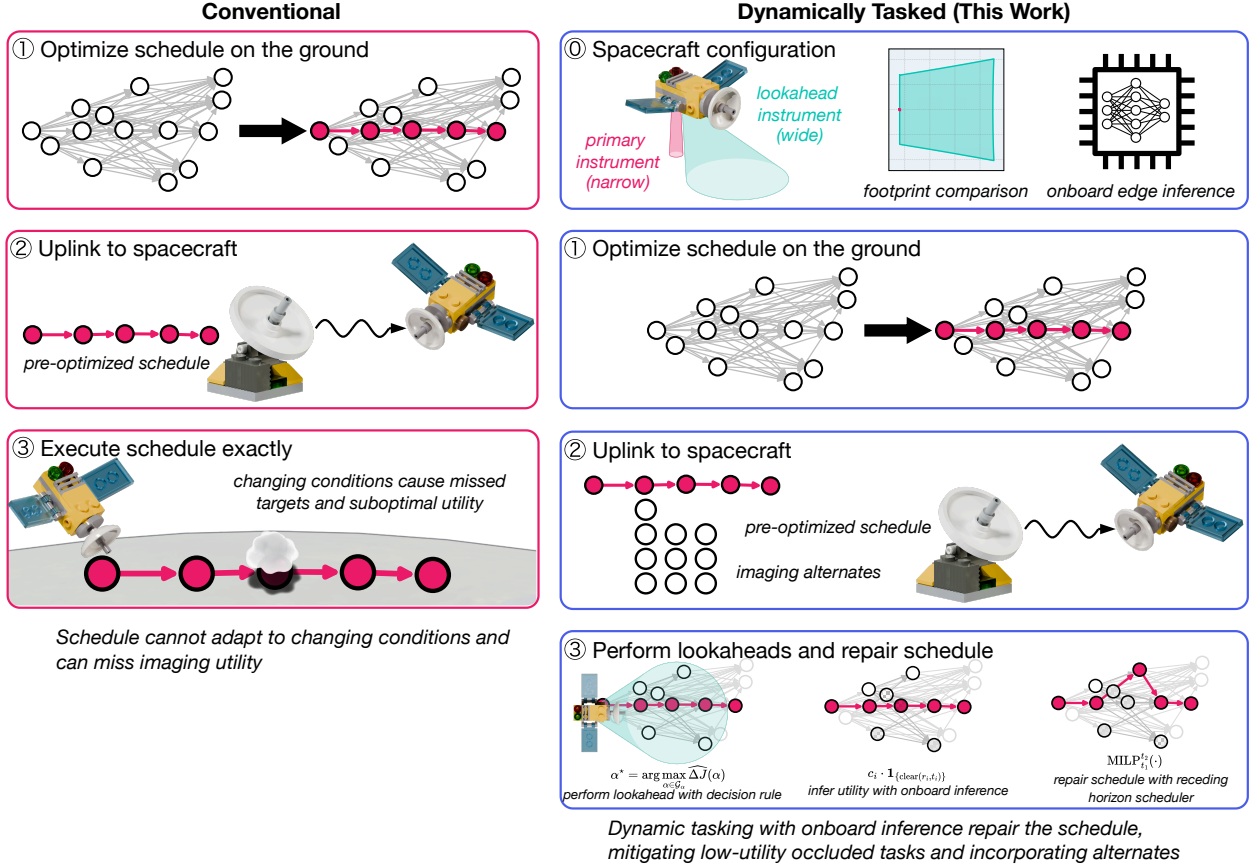


Figure 1: Dynamic-tasking ConOps. **(left)** A fixed ground-optimized schedule can execute low-value collects after utility changes. **(right)** A body-fixed lookahead, onboard inference, and receding-horizon repair update the schedule before primary imaging.

Previous work searches directly over lookahead actions or fixes the observation strategy [2, 3]; we instead optimize lookahead pitch and observe only when the expected repair value is positive. Conventional execution and omniscient scheduling provide lower and upper reference policies, and their expected gap is the expected value of perfect information (EVPI) [4, 5]. We report the empirical fraction of the realized gap closed by each dynamic-tasking policy as an EVSI-like performance measure, but do not identify it with optimal expected value of sample information because the evaluated policies are not themselves proven optimal.

Viewed through this lens, the systems analysis defines which information-gathering actions are physically available, the value model asks whether an action can repay its local schedule burden, and the timing and pointing rules maximize that margin within a schedule gap.

We make the following contributions:

1. A **systems analysis** linking ConOps, body-fixed field of view, geometry, and agility to feasible information-gathering actions (§3).
2. A **two-anchor renewal heuristic** and monotone break-even gate for lookahead actions (§4.1).
3. A **receding-horizon scheduler** combining pitch search, closed-form timing, and local repair (§4.2–§4.3).

2 Background

EO scheduling. Greedy dispatch rules suit onboard use [6, 7, 8], while MILPs optimize known utility [9, 10, 11]. Graph formulations exploit sparse access feasibility for cheaper near-optimal solutions [12, 13, 14]; MDP, reinforcement-learning, and Monte Carlo tree search methods avoid the access-based formulation [15, 16, 17, 18, 19].

These methods decide how to optimize a known set of accesses, but the central obstacle for dynamic tasking is not only schedule optimization: it is deciding which uncertain future accesses are worth observing before the primary instrument commits.

Dynamic targeting and tasking. Dynamic *targeting* acquires information and creates targets within an overflight; dynamic *tasking* repairs an initial schedule over fixed requests. EO-1 pioneered onboard replanning and ML classification, although limited compute restricted it mainly to follow-up observations [20, 21, 22]. GOSAT-2 now redirects its primary instrument from clouds with a fold mirror [23], and CogniSAT-6 demonstrated CNN-based autonomous tasking on a 6U CubeSat [24].

Onboard replanning has used anytime planners, timeline fragments, pointing heuristics, constellation-aware MILPs, and MDPs [25, 26, 27, 28, 11, 3]. The closest comparator is Kangaslahti et al. [2], which searches lookahead choices under slew and changing-utility constraints. Its different observation model lacks our body-fixed geometry, pitch-dependent support, and anchor-bounded repair, precluding a direct numerical comparison. These methods establish that onboard adaptation is possible, but generally search directly over sensing actions or assume a fixed observation strategy rather than first deriving the lookahead decision space from spacecraft constraints. The remaining gap is a computationally lightweight value model for body-fixed lookahead that connects geometry and agility to schedule-level benefit and formal timing guarantees. We address that gap and compare against conventional and omniscient bounds plus a greedy policy in the same simulator.

Onboard compute and agility. EO-1’s ~ 4 MIPS shared processor could not analyze lookahead imagery within an overflight [20, 21]; recent COTS VPU/GPU-class processors can [29, 30, 31, 24, 32, 33].

Agility determines whether lookahead information remains actionable [14, 34]. Representative platforms offer $\pm 45^\circ$ off-nadir and complete an extreme slew in about 25–40 s; Appendix A summarizes public specifications, and §3.3 derives the coupled constraints.

Sequential information acquisition. Dynamic tasking is sequential information acquisition: the spacecraft spends resources to reveal task values before committing its primary instrument. Related robotics methods value uncertainty reduction or coverage under travel, time, or energy constraints [35, 36]; inspection and probing models value information but usually assume separable items [37, 38]. Here a revealed target matters only if it fits a feasible schedule, coupling information value to agility and repair.

3 Systems Analysis

The onboard algorithm is derived from the flight-system constraints. Rather than solving an adaptive stochastic scheduling problem with continuous lookahead times, continuous pointing angles, unknown cloud states, and a combinatorial imaging schedule, we first use scheduling structure, spacecraft geometry, and agility limits to collapse dynamic tasking into local decisions made in schedule gaps. For each gap, the spacecraft asks whether one lookahead maneuver is physically meaningful and, if so, which pitch angles are feasible. The lookahead decision method then estimates whether any feasible maneuver has positive schedule-level value.

3.1 Baseline Scheduling Model

Orbital parameters map requests in $\mathcal{R}$ to accesses $a_i = (r_i, t_i, \theta_i)$, recording request, overpass time, and off-nadir roll [14]. Point, strip, and area requests are represented as points after standard tessellation or decomposition [14, 39]. The scheduling problem is to select a subset $\sigma \subseteq \mathcal{A}$ maximizing total utility subject to agility constraints:

$$\max_{\sigma} \sum_{a_i \in \sigma} u_i, \quad \text{s.t.} \quad t_j - t_i \geq t_{\text{slew}}(\theta_j - \theta_i) \quad \forall a_i, a_j \in \sigma, t_j > t_i, \quad (1)$$

where $t_{\text{slew}}(\Delta\theta) = t_s + \beta\sqrt{|\Delta\theta|}$ is a bang-bang model with settling time t_s and rate parameter β . A mixed-integer linear program (MILP) solves Equation 1 with an added constraint preventing repeated collection of a request.

3.2 Dynamic Tasking Formulation

We modulate each intrinsic access value c_i by cloud state:

$$u_i = c_i \cdot \mathbf{1}_{\{\text{clear}(r_i, t_i)\}}. \quad (2)$$

We study unit utility, $c_i = 1$ [40]. A *lookahead observation* reveals cloud states unknown during ground planning, after which the schedule is repaired with updated utilities.

Lookahead utility is the *advantage* of revealed, insertable clear accesses minus the *opportunity cost* of slew time, reducing the problem to local decision analysis.

We analyze the body-fixed onboard-lookahead case because it is tightly coupled to spacecraft agility: the same bus that images targets must also slew to acquire information about them. The field of view, horizon lead time, and slew time therefore directly determine the feasible onboard decision space. This coupling is weaker for sensor webs, leader-follower constellations, external meteorology, radar, and steerable instruments, which instead shift the trade toward communication latency, sensor cadence, and data-fusion assumptions [41, 42, 43, 44, 45].

3.3 Lookahead Geometry and Agility Constraints

For sizing, the lower vertical detector edge points toward nadir at zero body roll and pitch, so the boresight pitch is $b = \Theta_v/2$ for full vertical FoV Θ_v . We require the lookahead footprint to intersect each lateral boundary of the primary-imager field of regard at an actionable point. This is an edge-contact requirement: each boundary must be reached somewhere with enough lead time for the subsequent maneuver.

Lead time and vertical FoV. For a circular orbit with $a = R_E + h$ and $n = \sqrt{\mu/a^3}$, a visible point at signed along-track central angle ψ has lead time $t_{\text{lead}} = \psi/n$. The horizon bounds this lead by

$$t_{\text{horizon}} = \frac{1}{n} \arccos\left(\frac{R_E}{a}\right), \quad (3)$$

which is approximately 305 s at 400 km altitude. At zero body roll and pitch, the upper vertical detector edge points forward by Θ_v ; increasing Θ_v therefore extends the available lead, up to the limb (Figure 2). A contact is actionable only when $t_{\text{lead}} \geq t_{\text{req}}$, where t_{req} includes the applicable repointing, settling, acquisition, and inference times. Appendix B.1 gives the spherical ray intersection and detector projection.

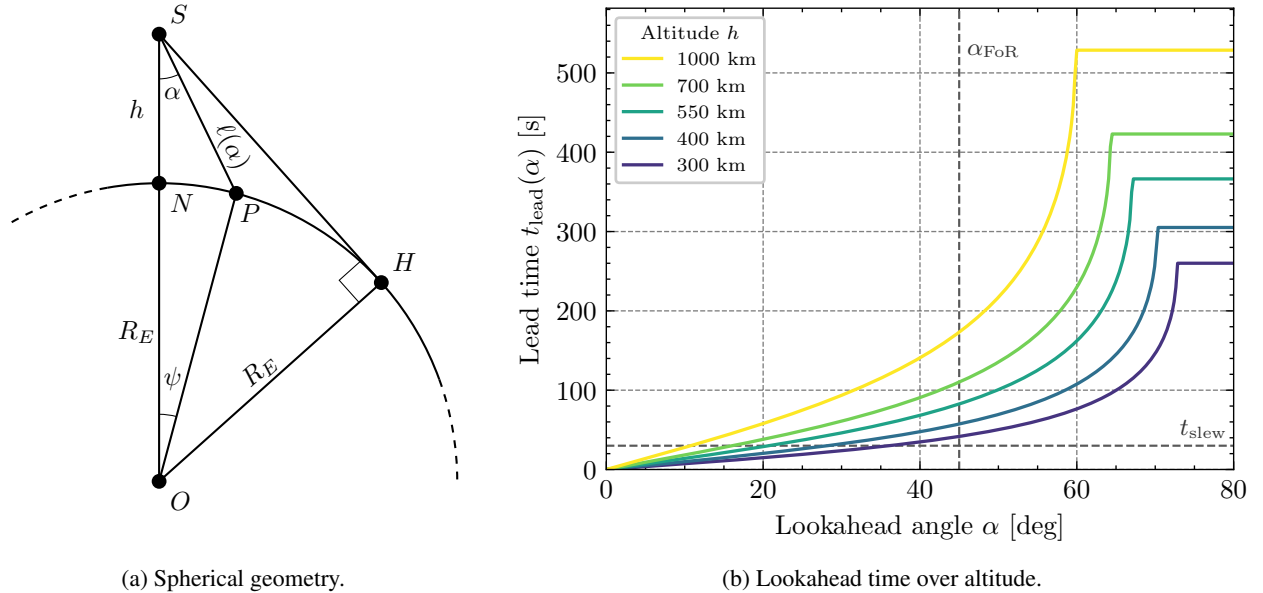


Figure 2: Lookahead geometry and timing. **(a)** Spacecraft S , nadir N , horizon tangent H , and target P at pitch α ; ψ is the signed along-track lead angle and ρ the tangent-plane projection of NP . **(b)** Lead time at representative LEO altitudes; dashed guides illustrate a pointing limit and a representative slew duration.

Detector bounds. Let δ , η , and d be a target's horizontal offset, vertical offset, and depth in the lookahead camera frame. Normalized detector coordinates are $u = \delta/d$ and $v = \eta/d$. For $d > 0$, the rectangular detector contains the target when

$$|u| \leq \tan \frac{\Theta_h}{2}, \quad |v| \leq \tan \frac{\Theta_v}{2}, \quad (4)$$

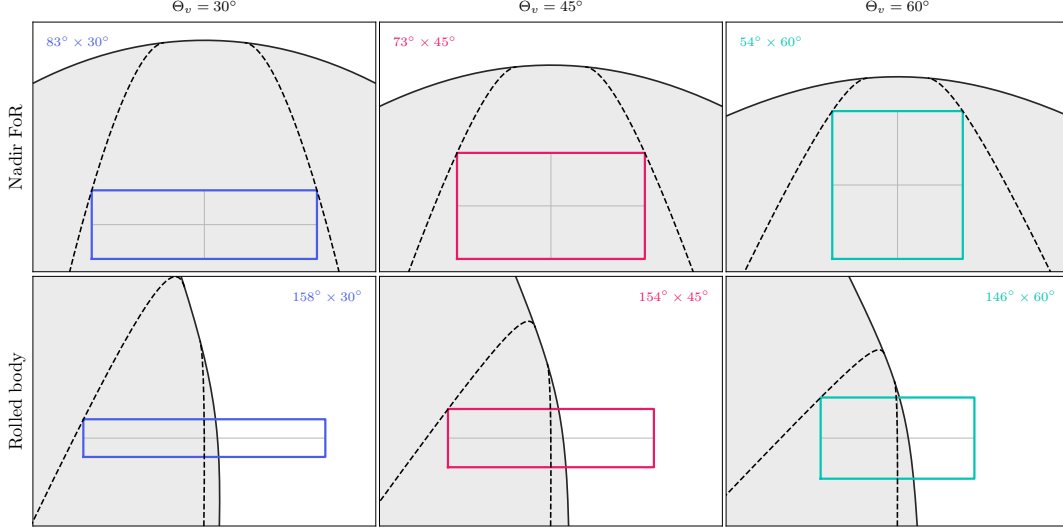


Figure 3: Common-focal detector limits for vertical FoVs of 30° , 45° , and 60° . Dashed curves mark primary-imager FoR, black curves the limb, and shading Earth. The top row uses nadir at $\theta_{\text{FoR}} = 45^\circ$; the bottom uses the native rolled frame at $\theta_{\text{FoR}} = 40^\circ$, exposing the near- 180° body-fixed width.

where Θ_h is full horizontal FoV. Thus horizontal sizing asks how wide the detector must be to reach both projected FoR boundaries (Figure 3).

Horizontal sizing. For body roll φ , let $\mathcal{D}_\sigma(\varphi; t_{\text{req}})$ contain the Earth-visible points on FoR edge $\sigma \in \{-1, +1\}$ that are front-facing, satisfy the vertical bound in Equation 4, and have sufficient lead time. Parameterizing each edge by along-track position x , the required horizontal FoV is

$$\Theta_{h,\min}(\varphi; t_{\text{req}}) = 2 \arctan \left[\max_{\sigma \in \{-1, +1\}} \inf_{x \in \mathcal{D}_\sigma(\varphi; t_{\text{req}})} |u_\sigma(x; \varphi)| \right]. \quad (5)$$

For each edge, we find the contact requiring the smallest width, then size for the harder edge. If either admissible set is empty, that attitude is infeasible. We evaluate nadir ($\varphi = 0$) and maximum roll ($\varphi = \theta_{\text{FoR}}$) separately; an instrument required to support both takes the larger width. These are two design attitudes, rather than a guarantee over intermediate rolls.

Flat-Earth sizing rule. At nadir, a flat-Earth approximation makes the trade explicit. A primary-imager FoR half-angle θ_{FoR} places the lateral boundaries at $y = \pm h \tan \theta_{\text{FoR}}$. The upper detector edge reaches $x = h \tan \Theta_v$, with camera depth $d = x \sin b + h \cos b$. Substitution in Equation 4 gives

$$\tan \frac{\Theta_{h,\min}^{(N)}}{2} \approx \frac{h \tan \theta_{\text{FoR}}}{h \tan \Theta_v \sin b + h \cos b} = \frac{\tan \theta_{\text{FoR}} \cos \Theta_v}{\cos(\Theta_v/2)}, \quad 0 < \Theta_v < 90^\circ. \quad (6)$$

The rule applies when this contact has sufficient lead time. A larger vertical FoV reaches farther ahead, where the lateral boundaries occupy less horizontal detector width. For $\theta_{\text{FoR}} = 45^\circ$, vertical FoVs of 45° and 60° require approximately 74.9° and 60° horizontal FoV; spherical projection at 400 km gives 73.0° and 54.4° . Near the limb, especially at large roll, we use the spherical calculation in Appendix B.2.

Instrument configurations. The $60^\circ \times 60^\circ$ design in Table 1 meets the geometric nadir edge-contact requirement, while the $45^\circ \times 45^\circ$ design requires reorientation to reach both lateral boundaries. At maximum roll with $\theta_{\text{FoR}} = 45^\circ$, neither design reaches both edges (Figure 4), motivating active body reorientation. Both designs use the pitch-search policy in the scheduling experiments; maneuver feasibility is evaluated in each schedule gap.

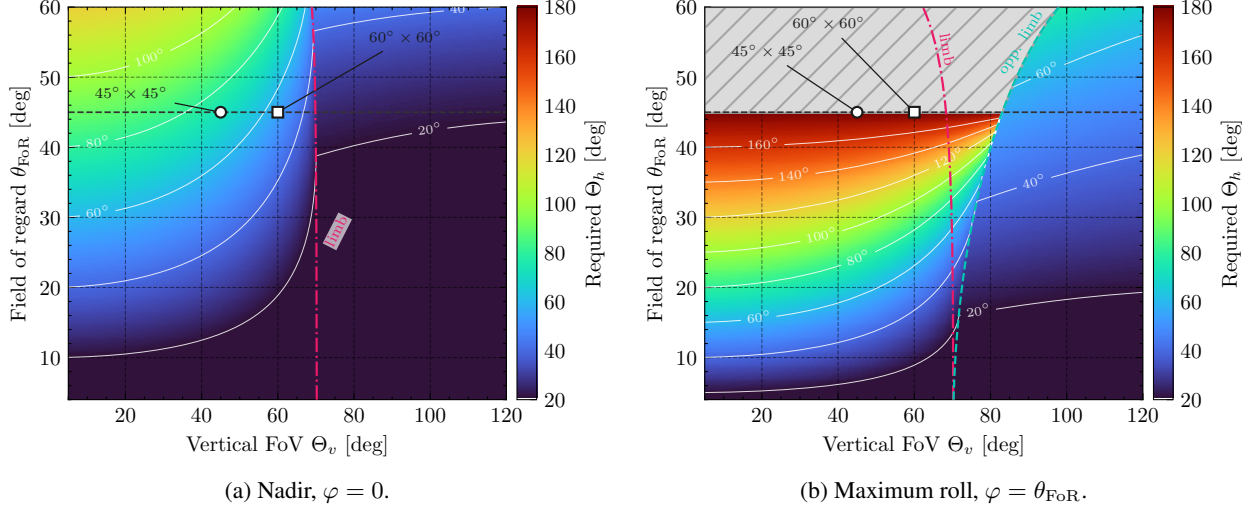


Figure 4: Geometric edge-contact requirements at $h = 400$ km from Equation 5 with $t_{\text{req}} = 0$. Color gives the horizontal FoV needed to touch both lateral boundaries at the indicated attitude. Magenta curves mark limb contact; cyan marks opposite-edge limb contact; hatching marks no admissible contact with both edges. Markers locate the two instrument designs at a primary FoR of $\pm 45^\circ$. A positive maneuver-time requirement can further restrict contact.

Table 1: Lookahead instrument configurations.

Case	Vertical FoV	Horizontal FoV	Boresight pitch
Wide	60°	60°	30°
Compact	45°	45°	22.5°

3.4 Agility Limits

The same systems constraints bracket any dynamic-tasking policy. With strictly positive lead and vanishing maneuver, acquisition, and inference time, full advance revelation recovers the omniscient MILP. Feasible lookahead sets shrink monotonically with slew time; once no schedule gap admits a round trip, dynamic tasking reduces to conventional execution.

This systems analysis reduces the onboard policy to one candidate lookahead action per schedule gap, searched over a discrete grid of feasible pitch angles. The remaining question is how to value each candidate angle.

4 Lookahead Decision Method

We approximate schedule repair inside a bounded lookahead window using the following four assumptions:

- (A1) $\tau_{ij} \approx g \quad \forall (i, j)$ inside the local repair window, constant effective spacing;
- (A2) $t_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}(T_1, T_2)$, exchangeable access times;
- (A3) $X_i \stackrel{\text{i.i.d.}}{\sim} \text{Bern}(p)$, i.i.d. cloud states;
- (A4) $c_i \approx \bar{c}$, locally equal intrinsic utility. (7)

(A1) compresses pairwise slews into an effective dead time; (A2) equates fixed-count uniform times with Poisson arrivals conditioned on their count. (A3)–(A4) express value in expected clear-image equivalents; the repair MILP can still use heterogeneous priorities after acceptance.

4.1 Two-Anchor Lookahead Value Model

At pitch α relative to rest angle α_0 , camera projection gives the unobserved, visible accesses no earlier than the next scheduled image: $\mathcal{V}_\alpha$, with $N_{\text{obs}}(\alpha) = |\mathcal{V}_\alpha|$. Before scoring pitches, each gap fixes t_L as the previous scheduled image

(or window start) and t_R as the first scheduled image after the largest feasible detector support. Let

$$L_{\text{anc}} = t_R - t_L \quad (8)$$

be the common anchor interval, translated to $[0, L_{\text{anc}}]$ for every pitch. The detector need not span it. For pitch α , let

$$0 \leq \ell_\alpha < r_\alpha \leq L_{\text{anc}}, \quad W_\alpha = r_\alpha - \ell_\alpha \quad (9)$$

be detector support after latest-feasible timing, clipped near the next scheduled image and far at the horizon. Let N_{sched} count scheduled tasks in (t_L, t_R) and N_{missed} those forfeited during slew. The net lookahead utility is

$$\widehat{\Delta J} = \underbrace{\hat{J}^+(\mathcal{V}_\alpha)}_{\text{advantage}} - \underbrace{\hat{J}^-(\mathcal{V}_\alpha)}_{\text{opportunity cost}}. \quad (10)$$

Opportunity cost. Each forfeited task has expected clear utility $p\bar{c}$, so

$$\hat{J}^- = p \cdot \bar{c} \cdot N_{\text{missed}}, \quad (11)$$

where $p = P(\text{clear})$ and $\bar{c}$ is mean intrinsic utility. The timing rule in §4.2 normally makes $N_{\text{missed}} = 0$; we retain it for calibration and alternative timing.

Poissonized chain model. An observed clear access has value only if insertable between its scheduled anchors. Within the revealed support, prior thinning gives clear-candidate density

$$\lambda_c(\alpha) = \frac{pN_{\text{obs}}(\alpha)}{W_\alpha}. \quad (12)$$

The stochastic approximation exploits a standard symmetry: conditional on observing N_{obs} arrivals in an interval, the unordered times of a homogeneous Poisson process are i.i.d. uniform [46, 47]. We therefore represent the fixed visible candidates by a process conditioned on its count, then approximate the unknown clear subset by an unconditioned thinned process of rate λ_c . Compressing pairwise slews into dead time g turns local repair into a one-dimensional packing problem. In body-fixed experiments, g_{phys} is the conservative worst-case slew across the primary-imager FoR; tighter local or angle-dependent bounds can replace it. With the left and right schedule anchors translated to 0 and L_{anc} , an insertable clear chain must satisfy

$$t_1 \geq g, \quad t_{i+1} - t_i \geq g, \quad L_{\text{anc}} - t_k \geq g. \quad (13)$$

The anchors remove support lying within g of either boundary. Define

$$c_L(\alpha, g) = \max(0, g - \ell_\alpha), \quad c_R(\alpha, g) = \max(0, g - (L_{\text{anc}} - r_\alpha)), \quad (14)$$

and the usable support length

$$U_\alpha(g) = W_\alpha - c_L(\alpha, g) - c_R(\alpha, g). \quad (15)$$

After accepting a chain member, the next requires clearance g plus an exponential wait of mean $1/\lambda_c$ [48, 49]:

$$\mathbb{E}[R] = g + \frac{1}{\lambda_c}, \quad (16)$$

where R is repaired-chain spacing.

For finite support $[\ell_\alpha, r_\alpha]$, the probability of selecting at least k internal points with gaps at least g is

$$G_\alpha(k; g) = \frac{\gamma(k, \lambda_c(U_\alpha(g) - (k-1)g))}{\Gamma(k)}, \quad U_\alpha(g) \geq (k-1)g, \quad (17)$$

with $G_\alpha(k; g) = 0$ when $U_\alpha(g) < (k-1)g$. The corresponding expected number of insertable internal points is

$$\widehat{K}_{2a}(\alpha, g) = \sum_{k \geq 1} G_\alpha(k; g). \quad (18)$$

For full support, $c_L = c_R = g$ and Equation 17 has residual length $L_{\text{anc}} - (k+1)g$. The finite-window advantage is

$$\hat{J}_{2a}^+ = \bar{c} \widehat{K}_{2a}(\alpha, g), \quad (19)$$

The onboard gate can evaluate $\widehat{K}_{2a}(\alpha, g_{\text{phys}})$ directly; the sum is finite, monotone in g , and cheap for short windows.

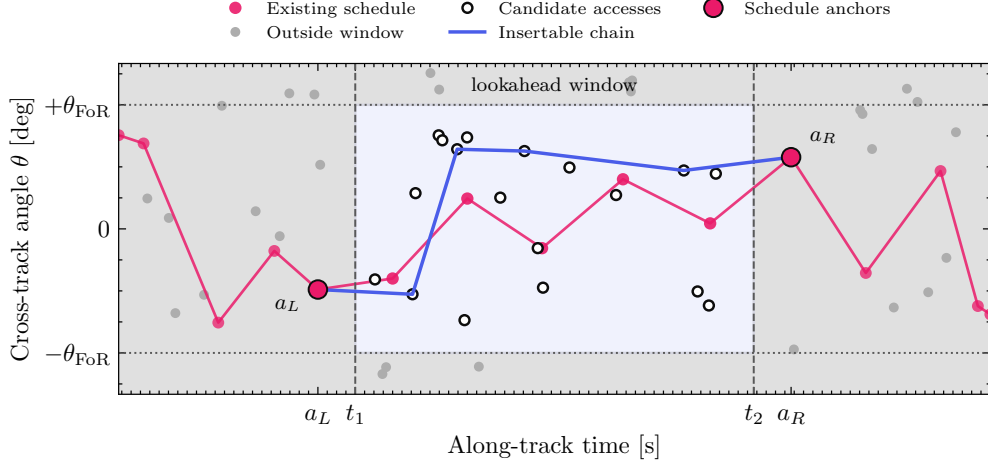


Figure 5: Two-anchor value window. Scheduled anchors bound local repair; gray regions are not actionable. A lookahead must add more expected insertable utility than the local replacement burden.

Break-even approximation. For a closed-form threshold, we approximate the finite-window chain by its long-run renewal rate on the same usable detector support:

$$\hat{K}_b(\alpha, g) \approx \frac{U_\alpha(g)}{g + 1/\lambda_c(\alpha)} = \frac{\lambda_c(\alpha)U_\alpha(g)}{1 + \lambda_c(\alpha)g}. \quad (20)$$

This average packing rate increases with candidate density and decreases monotonically with dead time, enabling an algebraic threshold. We use the finite two-anchor form as the accurate estimator and $\hat{K}_b$ as the break-even approximation. Appendix B.3 derives Equation 17 and discusses the residual bias.

Now write the net advantage in units of average task utility:

$$\frac{\widehat{\Delta J}_{2a}}{\bar{c}} = \hat{K}_{2a}(\alpha, g) - \left(p N_{\text{sched}} + \frac{\hat{j}^-}{\bar{c}} \right). \quad (21)$$

We define

$$B = p N_{\text{sched}} + \frac{\hat{j}^-}{\bar{c}}, \quad (22)$$

as the local replacement burden measured in clear-image equivalents. The first term is the expected utility already present in the scheduled anchor interval and therefore represents what the repaired schedule must preserve. The second converts maneuver-induced loss into the same units. Thus B is not a cloud or geometry parameter by itself; it is the number of insertable clear-image equivalents a lookahead must produce before it creates net schedule value. Fixing B across each pitch grid avoids boundary discontinuities. It is conservative for partial detector support because repair may retain scheduled tasks outside that support. The finite-window gate is $\hat{K}_{2a}(\alpha, g_{\text{phys}}) > B$; the engineering threshold solves $\hat{K}_b(\alpha, g_{\text{break}}) = B$. Equivalently,

$$g_{\text{break}}(\alpha) = \sup \left\{ g \geq 0 : \frac{\lambda_c(\alpha)U_\alpha(g)}{1 + \lambda_c(\alpha)g} > B \right\}. \quad (23)$$

If $\lambda_c(\alpha)U_\alpha(0) \leq B$, even zero dead time cannot beat the baseline. For full anchor support, Equation 23 becomes

$$g_{\text{break}} \approx \frac{L_{\text{anc}}(pN_{\text{obs}} - B)}{pN_{\text{obs}}(2 + B)}, \quad pN_{\text{obs}} > B. \quad (24)$$

Monotonicity in g makes the gate

$$\widehat{\Delta J} > 0 \iff g_{\text{phys}} < g_{\text{break}}, \quad (25)$$

up to the approximation in Equation 20.

g_{break} is not a replacement for the spacecraft slew time. It is the maximum effective dead time that the current observation window can tolerate before the expected value of looking falls to zero. The policy therefore compares this

schedule-derived threshold with the physical dead time g_{phys} ; with zero opportunity cost, the threshold increases with usable support and decreases with existing schedule utility.

Because projection and timing change observed sets, support, and clearances, each pitch has margin

$$M(\alpha) = g_{\text{break}}(\alpha) - g_{\text{phys}}(\alpha), \quad \alpha^\dagger = \arg \max_{\alpha \in \mathcal{G}_\alpha^{\text{feas}}} M(\alpha), \quad (26)$$

and acceptance requires $M(\alpha^\dagger) > 0$. The two-anchor policy instead maximizes $\hat{K}_{2a}(\alpha, g_{\text{phys}}) - B$ on the same support and baseline. Body-fixed scoring uses $g_{\text{phys}} = t_{\text{slew}}(\theta_{\text{FoR}})$ to tolerate worst-case cross-track slews. Angle-dependent dead time can replace it; inadequate resolution or classifier performance can exclude pitches from $\mathcal{G}_\alpha^{\text{feas}}$.

4.2 Lookahead Sensing Policy: Timing and Pitch

Each gap selects no action or an analytic initiation time plus one-dimensional pitch.

Latest feasible lookahead timing. Let $[t_{\text{prev}} + t_s, t_{\text{next}}]$ be a maximal idle gap. Timing trades two opposing effects: a later observation sees farther into the future and can reveal more clear candidates, but a round trip that extends past t_{next} begins to displace the next scheduled image. We admit at most one round trip and require it to finish by t_{next} , preserving both anchors and eliminating that opportunity cost. Within this feasible set, later observations weakly enlarge the revealed set, so the latest feasible start is

$$t^*(\alpha) = t_{\text{next}} - t_{\text{man}}(\alpha), \quad (27)$$

where $t_{\text{man}}(\alpha) = 2(t_{\text{slew}}(\alpha - \alpha_0) - t_s)$ is the round-trip maneuver time for pitch angle α and the *actionable* observation window has length

$$L_{\text{act}}(t) = t + t_{\text{horizon}} - t_{\text{next}}. \quad (28)$$

Proposition 1 (Latest Feasible Lookahead Timing). *Fix a feasible pitch angle α and restrict admissible lookahead maneuvers to round trips fully contained in the current schedule gap. Under the fixed gap-level baseline used by the policy, $t = t^*(\alpha)$ maximizes the estimated net advantage $\widehat{\Delta J}$ over all admissible initiation times. At this time $\hat{J}^- = 0$, and the value gate is evaluated on the clipped detector support inside the fixed anchor interval.*

A lookahead is feasible when $t_{\text{next}} - t_{\text{prev}} - t_s \geq t_{\text{man}}(\alpha)$; Appendix B proves the result.

Break-even pointing search. Pitch trades access count against maneuver time: steeper views reveal more accesses but shorten actionability. The policy evaluates Equation 23 on the feasible grid with fixed gap-level baseline, chooses $\alpha^\dagger$ from Equation 26 if $M(\alpha^\dagger) > 0$, and otherwise does not look.

4.3 Schedule Repair

After a positive-value lookahead, receding-horizon re-optimization repairs only the affected segment [50, 11, 10]. The nearest scheduled tasks before and after the observed window anchor a local MILP over the pre-optimized schedule and alternate-access queue. Observed-cloudy accesses are removed; observed-clear and unobserved accesses remain, the latter at prior utility pc_i . The repaired segment is then spliced into the schedule.

Algorithm 1 gives the break-even loop; experiments also test direct two-anchor scoring. Both assume correct, zero-latency classification within the lookahead FoV.

5 Experiments

5.1 Experiment Setup

All experiments draw from 10 000 globally distributed populated-place requests. Scheduling uses a 24 h window, circular 400 km orbit, and Table 1 instruments. Lookaheads are evaluated at imaging time, tying cloud state and insertion feasibility to acquisition epoch. Unless noted, the BCM prior is $p = 0.34$.

We compare four levels of information use. The *conventional* policy executes the ground-planned schedule under realized cloud truth, while the *omniscient* MILP re-optimizes after every access state is revealed and serves as an upper reference. The *greedy* lookahead policy selects the pitch revealing the largest number of unobserved accesses and looks whenever a nonempty view exists, without a schedule-value gate. The break-even and two-anchor policies look only when their respective value estimates are positive.

Algorithm 1 Dynamic tasking control loop

Require: initial schedule σ_0 , alternates queue $\mathcal{A}_{\text{alt}}$, angle grid $\mathcal{G}_\alpha$, cloud prior p

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1:  $\sigma \leftarrow \sigma_0$ 
2: for each schedule gap  $[t_{\text{prev}} + t_s, t_{\text{next}}]$  in  $\sigma$  do
3:    $\alpha^\dagger \leftarrow \arg \max_{\alpha \in \mathcal{G}_\alpha^{\text{feas}}} M(\alpha)$  ▷ break-even margin in Equation 26
4:   if  $M(\alpha^\dagger) > 0$  and  $t_{\text{next}} - t_{\text{prev}} - t_s \geq t_{\text{man}}(\alpha^\dagger)$  then
5:     at  $t^*(\alpha^\dagger) = t_{\text{next}} - t_{\text{man}}(\alpha^\dagger)$ : slew to  $\alpha^\dagger$ , observe, slew back ▷ Prop. 1
6:     update  $\{X_i\}$  for accesses in the observed window
7:      $\sigma \leftarrow \text{MILP-REPAIR}(\sigma, \mathcal{A}_{\text{alt}}, \text{bounding accesses})$ 
8:   end if
9:   execute scheduled imaging in  $[t_{\text{prev}}, t_{\text{next}}]$ 
10: end for
    
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For policy π , we report the realized conventional-to-omniscient gap fraction

$$f_{\text{closed}}(\pi) = \frac{J_\pi - J_{\text{conv}}}{J_{\text{omni}} - J_{\text{conv}}}, \quad (29)$$

alongside absolute clear-image utility. Lookahead use is the number of lookahead actions divided by the number of accesses in the initial conventional schedule.

5.2 Experiment 1: Heuristic Calibration

We calibrate two-anchor advantage $\hat{K}_{2a}(g_{\text{phys}}) - B$ and its break-even surrogate $\hat{K}_b(g_{\text{phys}}) - B$ against paired Monte Carlo local insertion advantage in sampled IID repair windows. The corresponding g_{break} and margin $M = g_{\text{break}} - g_{\text{phys}}$ remain gate diagnostics rather than estimates of advantage itself.

The sampler varies candidate density, cloud probability, physical spacing, and missed-task burden without conditioning on the sign of either heuristic. Each Monte Carlo value averages IID Bernoulli cloud states at fixed access times rather than running a full orbit-and-weather simulation. Thus this experiment isolates the exchangeable-access model; Experiments 2–3 add real geometry, pitch-dependent support, and schedule repair. Because $B = p(N_{\text{sched}} + N_{\text{missed}})$ is unbiased here, residual bias reflects the chain approximation.

We measure continuous agreement using the concordance correlation coefficient (CCC), which penalizes both dispersion and systematic offset from the one-to-one line. The two-anchor estimate attains CCC 0.992 with bias -0.025 , versus CCC 0.982 and bias -0.305 for break-even. Across 400 windows, two-anchor has 97.3% sign agreement, 2.1% false accepts among acceptances, and 3.6% false rejects among positive windows. Break-even has 92.3%, 1.8%, and 14.4%, respectively, confirming its conservatism. Against an oracle that accepts positive Monte Carlo advantage, mean regret is 0.005 versus 0.022 clear-image equivalents per window. 95% bootstrap intervals resample windows, propagate pointwise Monte Carlo error, and bound CCC, bias, and the least-squares line.

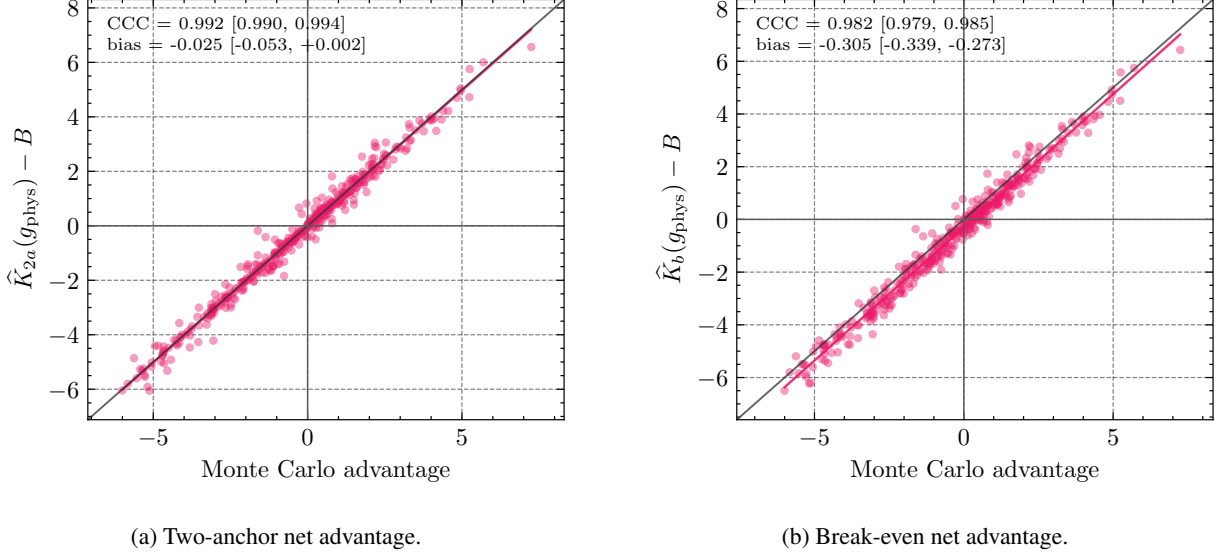


Figure 6: Net-advantage calibration on sampled IID repair windows. Each point is a synthetic window averaged over IID cloud draws. Panels compare Monte Carlo advantage with **(left)** two-anchor and **(right)** break-even estimates.

5.3 Experiment 2: IID End-to-End Scheduling

We test greedy, break-even, and two-anchor policies under IID Bernoulli truth. Each trial builds the conventional schedule, samples states with $p_{\text{clear}} = 0.34$, computes the omniscient bound, and replays policies on identical accesses and truth. Dates and RAAN phase match the BCM study; only cloud generation changes (Appendix E). Across 12 paired 24 h trials, we sweep $\Theta_h = \Theta_v \in \{45^\circ, 60^\circ\}$ and $t_{\text{slew}}(\theta_{\text{FoR}}) \in \{25, 35, 45\}$ s, scoring 64 pitches per gap. Across the sweep, greedy closes 58.1 % of the conventional-to-omniscient gap with 0.269 lookaheads per scheduled access. Break-even closes 78.2 % with 0.162 and two-anchor 79.0 % with 0.166: higher closure with about 40 % fewer actions. FoV dominates sensitivity. Enlarging it from $45^\circ \times 45^\circ$ to $60^\circ \times 60^\circ$ improves every agility cell; break-even gains 5.1–16.5 percentage points from 25 s to 45 s, with two-anchor similar. Greedy degrades with slew burden, while value gates avoid much of this loss by selecting fewer high-margin actions. At $60^\circ \times 60^\circ$, break-even/two-anchor close 80.6 %/80.9 % at 25 s and peak at 85.7 %/85.4 % at 45 s, while use rises only from about 0.16 to 0.20. These normalized fractions do not rank absolute utility: for $60^\circ \times 60^\circ$ break-even, mean clear images fall from 152.2 to 125.7 across those agility settings. Thus the policies schedule nearly identically; two-anchor estimates finite-window value more accurately, while break-even yields a dead-time threshold. Appendix C gives all cells.

5.4 Experiment 3: Realistic BCM Case Study

We repeat the sweep using geostationary Binary Cloud Mask truth in 12 paired 24 h windows. Appendix E gives trial starts and product identifiers. RAAN phase matching preserves the Earth-fixed ground track across dates; seasonal variation comes from illumination and clouds. Policies retain $p_{\text{clear}} = 0.34$, while truth uses the nearest product at each imaging time. No correlation model is fit: structure enters only through sampled masks. Paired policies share requests, accesses, conventional schedule, and truth.

Under BCM truth (Figure 8), break-even closes 77.6 % using 0.161 lookaheads per scheduled access, and two-anchor closes 79.0 % using 0.165. Greedy closes 43.6 % using 0.264. Relative to IID, break-even changes from 78.2 % to 77.6 % and two-anchor remains 79.0 %, whereas greedy drops from 58.1 % to 43.6 %. With dates, requests, geometry, and settings matched, scalar-prior screening therefore remains effective despite BCM correlation and coherent boundaries. The best normalized cell is $60^\circ \times 60^\circ$ at 45 s: break-even closes 85.6 % with 0.196 lookaheads per scheduled access, and two-anchor 86.0 % with 0.199. The small difference agrees with Experiment 1’s bias result. Appendix D reports absolute and normalized results; Appendix E reports timings.

5.5 Experiment 4: Overflight Case Study

The full-day sweeps hide within-pass behavior, so Figure 9 replays both gates on one access set, conventional schedule, camera model, and IID realization. The 30 min South America pass has 266 candidates and begins offshore; a zero-

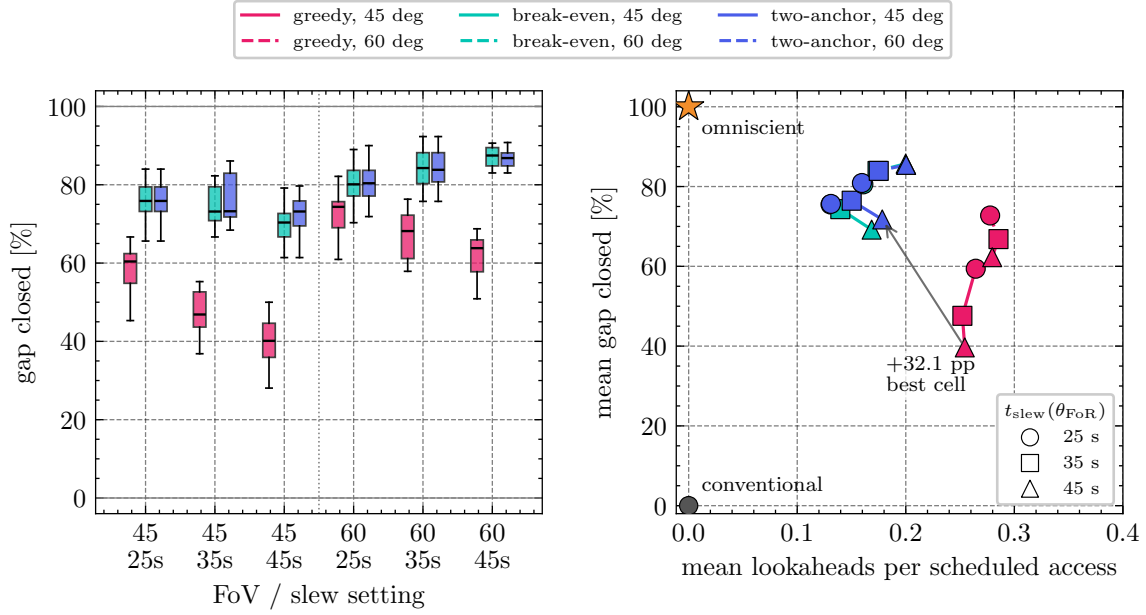


Figure 7: IID scheduling over 12 paired 24 h trials per FoV/agility cell. **Left:** gap fraction closed by trial. **Right:** mean closure versus lookaheads per scheduled access. Shape encodes agility; solid/dashed lines encode $45^\circ \times 45^\circ / 60^\circ \times 60^\circ$ FoV. The arrow marks the largest improvement over greedy; gray/orange mark conventional/omniscient bounds.

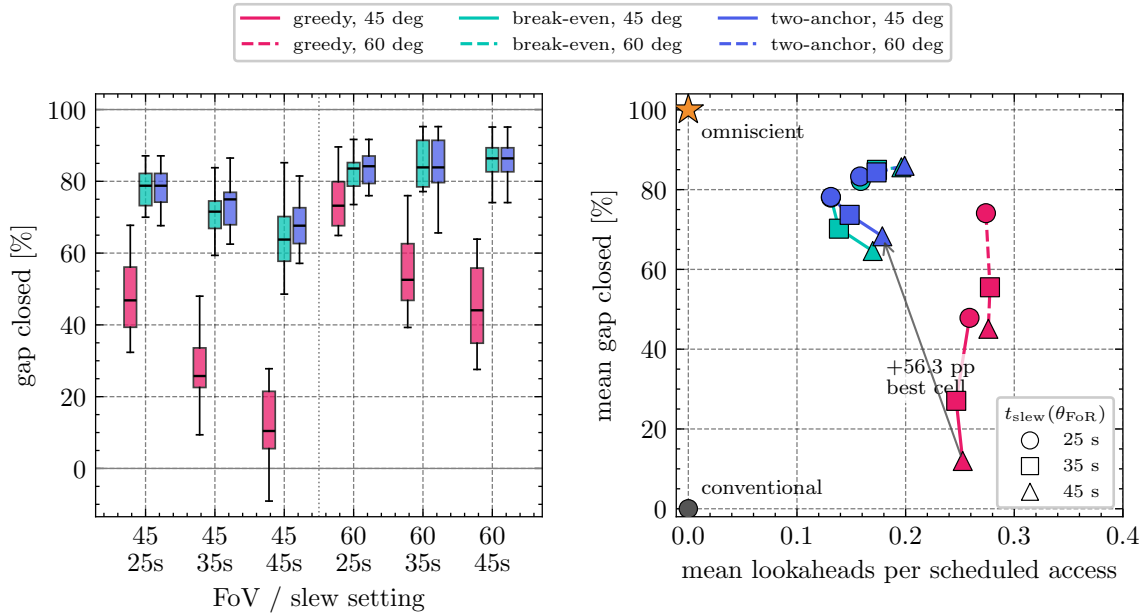


Figure 8: BCM scheduling over 12 paired 24 h trials per cell, with phase-matched RAAN and cloud masks sampled at imaging time. **Left:** gap fraction closed by trial. **Right:** mean closure versus lookaheads per scheduled access. Shape encodes agility; line style encodes FoV. The arrow marks the largest improvement over greedy; gray/orange mark conventional/omniscient bounds.

utility virtual anchor 2 min after its start exposes the initial reconnaissance gap before the first conventional image. Conventional execution schedules 36 real tasks and collects 7 clear images; omniscient scheduling collects 27. Both policies take 5 lookaheads, repair to 29 tasks, and collect 25 clear images.

The pass also exposes scalar pitch search’s geometric failure mode. A high-value access can enter the rectangular FoV laterally rather than through the modeled forward edge, yet projection includes it in the one-dimensional forward window. The gates still attain identical utility on different score scales.

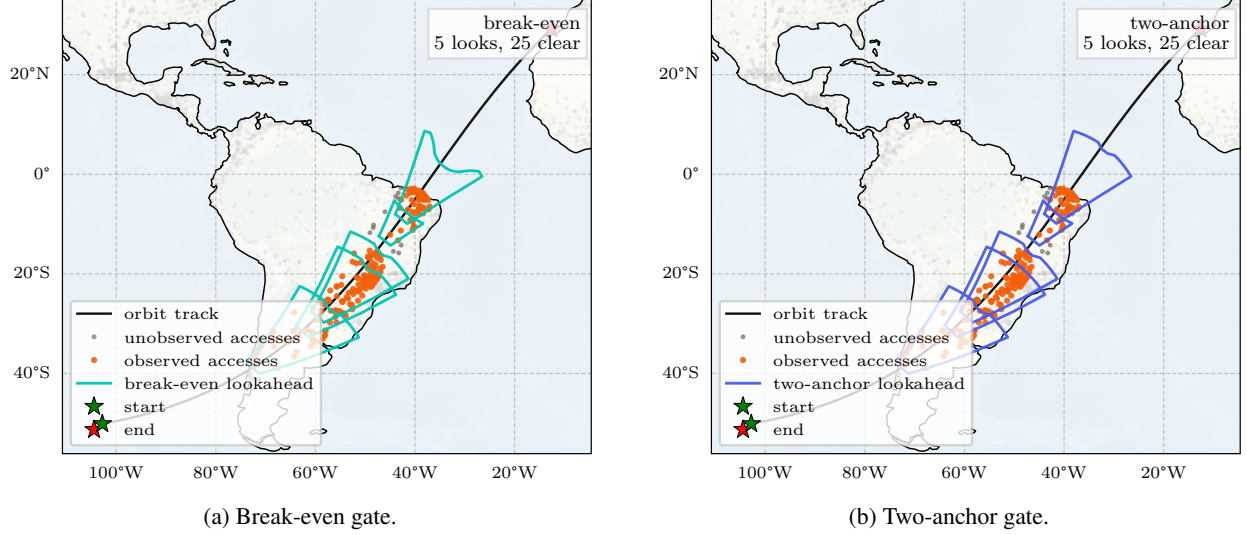


Figure 9: Brazil overpass. Orange dots are observed accesses, gray dots unobserved accesses, and colored outlines selected detector footprints.

6 Conclusions and Future Work

We derive a two-anchor renewal estimator, a computationally lightweight monotone break-even rule, and a closed-form latest initiation time with zero opportunity cost (Proposition 1). The resulting policy searches feasible pitches and observes only when estimated schedule improvement is positive. On sampled IID repair windows, the finite-window estimate closely matches paired Monte Carlo insertion values, while the break-even rule is conservative. In full 24 h scheduling trials, the value-based policies close roughly 78 %–79 % of the conventional-to-omniscient gap under both IID and realistic BCM cloud fields, while using about 40 % fewer lookaheads per scheduled access than a greedy lookahead baseline. The highest normalized cells reach about 86 % under both cloud models, although these fractions do not rank absolute utility across agility settings. In the overflight case, both value policies reach the same repaired clear-image count with the same lookahead count despite scoring views differently.

Limitations. The estimator assumes equal task utility (A4), i.i.d. cloud states (A3), and constant gap spacing (A1). BCM success under structured truth does not test prior mis-specification: $p_{\text{clear}} = 0.34$ is fixed, so richer beliefs and biased climatology remain open. Relaxing equal utility is especially important when Pareto-distributed values concentrate utility in few tasks.

Physical dead time is conservatively set by a worst-case cross-track slew across the full field of regard, not the realized local sequence. Flight software could recover margin with certified local bounds or angle-dependent costs.

Scalar pitch search reduces the two-dimensional footprint to along-track support. In the Brazil case, a high-value access enters through the rectangular FoV’s side rather than the chain model’s forward edge. Connected-component scoring or lateral-entry diagnostics could detect this mismatch.

The analysis also assumes perfect classification whenever an access is visible. False positives insert cloudy tasks; false negatives suppress alternatives; latency shrinks the actionable window; and high-pitch resolution loss can prevent classification. A mission implementation must therefore test precision, recall, latency, and resolution.

Finally, experiments compare a greedy baseline and conventional/omniscient bounds, not stronger anytime MILP/LNS, MCTS, MDP, or learned planners. Such comparisons must test whether the estimator still helps more capable repair methods.

Future work. Primary-instrument lookahead requires camera and cost models for imaging, retargeting, schedule degradation, and resolution-dependent classification. Radar, cross-satellite sensing, and meteorological data require corresponding visibility, latency, error, and maneuver models.

Algorithmic extensions include utility-weighted chains, correlated cloud and event beliefs, and multi-step lookahead within a gap. For constellations, shared beliefs could prevent redundant sensing, while ground planning could nominate lookahead opportunities before onboard repair.

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A Supporting System Data

Table 2: Agility specifications for representative tasked EO satellites, scaled to a 90° slew.

Platform	Launch	FoR	Manufacturer spec	Scaled 90° slew [s]
WorldView-1 [51]	2007	$\pm 45^\circ$	$2.5^\circ/\text{s}^2$, $4.5^\circ/\text{s}$ max	24.7
Pléiades 1A/1B [52]	2011	$\pm 47^\circ$	60° in 25 s (incl. settle)	37.5
SPOT-6 [53]	2012	$\pm 45^\circ$	30° in 14 s	42.0
Pléiades Neo [54]	2022	$\pm 47^\circ$	60° in 20 s	30.0

B Proofs and Derivations

B.1 Spherical Geometry and Detector Projection

Let $a = R_E + h$ and $n = \sqrt{\mu/a^3}$. A ray pitched ahead by α from nadir intersects the visible surface at slant range

$$\ell(\alpha) = a \cos \alpha - \sqrt{R_E^2 - a^2 \sin^2 \alpha}, \quad (30)$$

for $|\alpha| \leq \arcsin(R_E/a)$. Its signed along-track projection and central angle are

$$\rho(\alpha) = \ell(\alpha) \sin \alpha, \quad \psi(\alpha) = \text{atan2}(\rho(\alpha), a - \ell(\alpha) \cos \alpha). \quad (31)$$

Positive ψ denotes forward lead, with $t_{\text{lead}} = \psi/n$ in the circular-orbit sizing model.

Let x point forward and y cross-track in the plane tangent at nadir. We represent the two lateral FoR boundaries by $y = \sigma \rho_F$, where $\sigma \in \{-1, +1\}$ and $\rho_F = \rho(\theta_{\text{FoR}})$. With $R_F = \sqrt{R_E^2 - \rho_F^2}$, the spacecraft-to-surface depth and the visible along-track limit are

$$\begin{aligned} z(x) &= a - \sqrt{R_F^2 - x^2}, \\ x_{\text{limb}} &= \sqrt{R_E^2 [1 - (R_E/a)^2] - \rho_F^2}, \quad 0 \leq x \leq x_{\text{limb}}. \end{aligned} \quad (32)$$

An imaginary x_{limb} means the FoR edge is beyond the visible surface. Within this fixed-cross-track approximation, the forward lead angle is $\psi_F(x) = \text{atan2}(x, \sqrt{R_F^2 - x^2})$.

For boresight pitch $b = \Theta_v/2$ and body roll φ , rotating the line of sight into the camera frame gives

$$\begin{aligned} \delta_\sigma(x; \varphi) &= \sigma \rho_F \cos \varphi - z(x) \sin \varphi, \\ \eta_\sigma(x; \varphi) &= x \cos b - \sin b [z(x) \cos \varphi + \sigma \rho_F \sin \varphi], \\ d_\sigma(x; \varphi) &= x \sin b + \cos b [z(x) \cos \varphi + \sigma \rho_F \sin \varphi]. \end{aligned} \quad (33)$$

The normalized coordinates are $u_\sigma = \delta_\sigma/d_\sigma$ and $v_\sigma = \eta_\sigma/d_\sigma$. The admissible set used in Equation 5 is

$$\mathcal{D}_\sigma(\varphi; t_{\text{req}}) = \{x \in [0, x_{\text{limb}}] : d_\sigma > 0, \quad |v_\sigma| \leq \tan b, \quad \psi_F(x)/n \geq t_{\text{req}}\}. \quad (34)$$

The design maps set $t_{\text{req}} = 0$ to isolate geometric contact. Orbit propagation and actual access epochs supply the visibility and timing checks in the scheduler.

B.2 Evaluation of the Edge-Contact Requirement

Nadir contact. At $\varphi = 0$, the two edges are symmetric and $|u_\sigma| = \rho_F/(x \sin b + z \cos b)$ decreases with x . The minimum width is therefore attained at the farthest admissible point. For $\Theta_v < 90^\circ$, an upper-detector-edge contact obeys $x/z = \tan \Theta_v$, giving

$$x_{\text{edge}} = \sin \Theta_v \left[a \cos \Theta_v - \sqrt{R_F^2 - a^2 \sin^2 \Theta_v} \right]. \quad (35)$$

Use this contact when it is real and precedes the visible limb; otherwise use the limb if it satisfies the vertical detector bound. Equivalently, the transition occurs at $\Theta_v = \text{atan2}(x_{\text{limb}}, z(x_{\text{limb}}))$. With the retained point x_N , the spherical nadir requirement is

$$\Theta_{h, \min}^{(N)} = 2 \arctan \left[\frac{\rho_F}{x_N \sin b + z(x_N) \cos b} \right]. \quad (36)$$

A positive t_{req} additionally requires $\psi_F(x_N)/n \geq t_{\text{req}}$; otherwise no earlier point can meet the lead-time requirement.

Rolled contact. For the rolled design map, the implementation samples 5000 uniformly spaced positions in $[0, x_{\text{limb}}]$. For each edge, it projects the samples using Equation 33, retains those in $\mathcal{D}_\sigma(\varphi; 0)$, and takes the smallest $|u_\sigma|$. The larger of the two edge requirements gives the horizontal width. If either edge has no retained sample, the cell is marked infeasible at this sampling resolution. This one-dimensional numerical calculation avoids separate formulas for upper-edge, limb, and interior contacts. Figure 3 shows representative spherical projections.

B.3 Two-Anchor Renewal Estimator

Write $L = L_{\text{anc}}$, $[\ell, r] = [\ell_\alpha, r_\alpha]$, $\lambda = \lambda_c$, and $U = U_\alpha(g)$. After the boundary clearances in Equation 14–Equation 15, the first point can occur at usable-support start; each additional point needs clearance g plus an exponential wait. Under Poissonization [55, 48], placing k points requires

$$S_k = (k-1)g + \underbrace{\sum_{i=1}^k X_i}_{Y_k}, \quad (37)$$

where $Y_k \sim \Gamma(k, \lambda)$ [46]. Thus k Poisson arrivals must fit in $U - (k-1)g$. Poisson–Gamma duality with $\mu = \lambda[U - (k-1)g]$ gives Equation 17. Shape k counts exponential waits; anchor and internal clearances are deterministic, the two-anchor non-paralyzable (Type I) convention [49].

Residual bias. Equation 17 removes only support violating anchor clearances, eliminating the dominant one-anchor boundary bias. Remaining negative bias comes from replacing N fixed points by $\text{Poisson}(N)$: concavity of chain length $f(n)$ gives $\mathbb{E}[f(\text{Poisson}(N))] \leq f(N)$. An exact variant could average over $\text{Binomial}(N_{\text{obs}}, p)$; we retain Poissonization for monotonicity in g and algebraic pitch search. A de-Poissonization correction [55] adds numerical variance without improving RMSE.

B.4 Proof of Proposition 1 (Latest Feasible Lookahead Timing)

Decompose $\widehat{\Delta J}(t)/\bar{c} = E[K; \lambda_c(t), W(t), \ell(t), r(t), L_{\text{anc}}, g] - p N_{\text{sched}}(t) - \hat{J}^-(t)/\bar{c}$, where $E[K; \cdot]$ is the support-aware two-anchor chain-length estimator from Equation 18 applied to the clipped detector support $[\ell(t), r(t)]$ inside the fixed anchor interval. Over admissible times, $N_{\text{obs}}(t)$ and support end $r(t)$ are non-decreasing as the actionable window $[t_{\text{next}}, t + t_{\text{horizon}}]$ expands, so the chain estimate cannot decrease. The gap-level baseline is fixed, and every admissible trip ends by t_{next} , making $\hat{J}^- = 0$. Hence $\widehat{\Delta J}(t)$ is non-decreasing and maximized at $t^*(\alpha) = t_{\text{next}} - t_{\text{man}}(\alpha)$. $\square$

C IID Scheduling Numerical Summary

FoV [deg]	t_{slew} [s]	Policy	$\bar{J}_{\text{conv}}/\bar{J}_{\text{DT}}/\bar{J}_{\text{omni}}$	Gap closed [%]	Lookaheads/access
45	25	greedy	90.2/136.0/166.4	59.4 ± 7.2	0.264 ± 0.012
45	25	break-even	90.2/148.3/166.4	75.5 ± 5.5	0.131 ± 0.005
45	25	two-anchor	90.2/148.4/166.4	75.6 ± 5.3	0.131 ± 0.005
45	35	greedy	78.5/111.9/148.2	47.6 ± 5.9	0.252 ± 0.010
45	35	break-even	78.5/130.8/148.2	74.3 ± 5.4	0.140 ± 0.009
45	35	two-anchor	78.5/132.3/148.2	76.4 ± 6.5	0.150 ± 0.009
45	45	greedy	69.5/95.6/134.7	39.7 ± 6.5	0.254 ± 0.012
45	45	break-even	69.5/114.7/134.7	69.2 ± 6.7	0.169 ± 0.011
45	45	two-anchor	69.5/116.2/134.7	71.8 ± 6.7	0.178 ± 0.012
60	25	greedy	90.2/146.1/166.4	72.7 ± 5.4	0.278 ± 0.014
60	25	break-even	90.2/152.2/166.4	80.6 ± 5.6	0.161 ± 0.006
60	25	two-anchor	90.2/152.4/166.4	80.9 ± 5.5	0.160 ± 0.007
60	35	greedy	78.5/125.4/148.2	66.8 ± 6.0	0.285 ± 0.018
60	35	break-even	78.5/137.4/148.2	83.9 ± 5.3	0.174 ± 0.010
60	35	two-anchor	78.5/137.4/148.2	83.9 ± 5.1	0.175 ± 0.011
60	45	greedy	69.5/110.0/134.7	62.3 ± 5.5	0.280 ± 0.016
60	45	break-even	69.5/125.7/134.7	85.7 ± 5.5	0.200 ± 0.010
60	45	two-anchor	69.5/125.4/134.7	85.4 ± 5.0	0.200 ± 0.011

Table 3: Paired IID results from Figure 7: absolute clear-image means and mean $\pm$ SD gap closure and lookahead use.

D BCM Scheduling Numerical Summary

FoV [deg]	t_{slew} [s]	Policy	$\bar{J}_{\text{conv}}/\bar{J}_{\text{DT}}/\bar{J}_{\text{omni}}$	Gap closed [%]	Lookaheads/access
45	25	greedy	107.5/127.9/150.2	47.9 ± 11.2	0.259 ± 0.017
45	25	break-even	107.5/140.9/150.2	78.0 ± 5.8	0.131 ± 0.006
45	25	two-anchor	107.5/141.0/150.2	78.2 ± 6.1	0.131 ± 0.006
45	35	greedy	96.2/107.2/135.5	27.1 ± 11.1	0.247 ± 0.015
45	35	break-even	96.2/123.8/135.5	70.2 ± 9.6	0.139 ± 0.008
45	35	two-anchor	96.2/125.2/135.5	73.7 ± 7.3	0.148 ± 0.007
45	45	greedy	87.8/92.0/121.9	12.0 ± 11.6	0.253 ± 0.013
45	45	break-even	87.8/109.7/121.9	64.6 ± 11.3	0.170 ± 0.014
45	45	two-anchor	87.8/111.1/121.9	68.3 ± 8.0	0.179 ± 0.010
60	25	greedy	107.5/139.2/150.2	74.1 ± 7.6	0.274 ± 0.020
60	25	break-even	107.5/142.8/150.2	82.2 ± 5.6	0.159 ± 0.009
60	25	two-anchor	107.5/143.2/150.2	83.3 ± 4.9	0.158 ± 0.009
60	35	greedy	96.2/118.2/135.5	55.5 ± 11.7	0.278 ± 0.020
60	35	break-even	96.2/129.8/135.5	85.0 ± 6.9	0.174 ± 0.014
60	35	two-anchor	96.2/129.7/135.5	84.3 ± 8.5	0.173 ± 0.014
60	45	greedy	87.8/103.2/121.9	45.1 ± 11.9	0.276 ± 0.026
60	45	break-even	87.8/117.2/121.9	85.6 ± 6.1	0.196 ± 0.015
60	45	two-anchor	87.8/117.3/121.9	86.0 ± 5.7	0.199 ± 0.014

Table 4: Paired BCM results from Figure 8: absolute clear-image means and mean $\pm$ SD gap closure and lookahead use.

E BCM Acquisition Provenance and Runtime

The 12 trials begin at 00:00 UTC on 2025-01-01, 2025-02-02, 2025-03-01, 2025-04-01, 2025-05-01, 2025-06-01, 2025-07-01, 2025-08-01, 2025-09-01, 2025-10-01, 2025-10-10, and 2025-12-01. The two non-month-start dates replace month-start alternatives with incomplete coverage. Trials span 24 h; RAAN phase matching to January repeats the Earth-fixed ground track.

For each access, the implementation selects the nearest sub-satellite longitude, rounds imaging time to Table 5’s cadence, and verifies the product. The cache records source, exact time, and filename: `source_YYYY-MM-DD_HHMMSS.nc` for GOES/Himawari and `.grb` for Meteosat.

Archive	Spacecraft or service	Product or collection identifier	Cadence
NOAA	GOES West (GOES-18)	ABI-L2-ACMF	10 min
NOAA	GOES East (GOES-16 for Jan.–Apr.; GOES-19 for May–Dec.)	ABI-L2-ACMF	10 min
EUMETSAT	Meteosat Zero Degree Service	EO:EUM:DAT:MSG:CLM (MSGCLMK)	15 min
EUMETSAT	Meteosat Indian Ocean Data Coverage	EO:EUM:DAT:MSG:CLM-IODC (MSGCLMK)	15 min
NOAA	Himawari-9	AHI-L2-FLDK-Clouds objects containing AHI-CMSK	10 min

Table 5: Exact archive products used to construct the access-time BCM states. GOES East switches from GOES-16 to GOES-19 on 2025-04-07, between the April and May trial dates.

Runtime. An unoptimized Python paired 24 h replay over 64 pitches averaged 1.57 s/1.89 s for break-even/two-anchor under IID and 1.82 s/2.13 s under BCM. These desktop timings include bookkeeping and file I/O, so are not flight benchmarks; they only show low planning-horizon cost.